DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter-2023

Course: B. Tech **Semester :IV**

Branch : Electronics and Tele-communication Engineering

Subject Code & Name: BTBSC406_Y18 Numerical Methods & Computer Programming

Max Marks: 60 **Date:29/01/2024** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Find the absolute error & relative error in $\sqrt{6} + \sqrt{7} + \sqrt{8}$ correct to 4 significant digits.	**CO 02**	**6**
B)	Define errors. How many different types of errors occur while doing numerical computations? How they can be avoided?	**CO 01**	**6**
C)	If $X^2 - e^{-X} = 0$, find real root by Newton Raphson method	**CO 02**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	Solve the following equations $27x+6y-z = 85$ $X+y+54z = 110$ $6x+15y+2z = 72$ By using Guass Seidal method	**CO 03**	**6**
B)	Solve the following by Gauss elimination method $2x+y+z = 10$ $3x+2y+3z = 18$ $X+4y+9z = 16$	**CO 04**	**6**
C)	Solve the following equations by LU decomposition (Factorization method) $2x+3y+z = 9$ $X+2y+3z = 6$ $3x+y+2z = 8$	**CO 03**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	Apply Lagrange's formula inversely to obtain a root of the equation f(x)=0, given that f(30)= -30,f(34)= -13, f(38) =3, f(42) =18	**CO 05**	**6**

B)	The following values of y = f(x) are given X : 10 15 20 F(x) : 1754 2648 3564 Find the value of x for y =3000 by iterative method	**CO 03**	**6**
C)	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using i) Trapezoidal rule ii) Simpson's 1/3rd rule iii) Simpson's 3/8th rule	**CO 04**	**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	Apply Runge kutta 4th order method to find approximate value of y for x =0.2 in steps of 0.1, if dy/dx = x+y2 given that y=1 where x =0.		**6**
B)	Given dy/dx = y-x / y+x with initial condition y=1 at x=0, find y for x= 0.1 by using Euler's method		**6**
C)	Given the values X : 5 7 11 13 17 F(X) : 150 392 1452 2366 5202 Evaluate f(9) using Newton's divided difference formula.	**CO 05**	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	Explain following terms : i) Data hiding ii) Encapsulation iii) Polymorphism	**CO 06, CO 07**	**6**
B)	Explain the advantages of OOP? List features of OOP.	**CO 06**	**6**
C)	Differentiate between procedural and object oriented approach.	**CO 07**	**6**

***** End *****